

Sustainability in Car Manufacturing

The automotive industry has grown massively in recent decades, one of the largest industries in the world, and along with it the debate of environmental impact and sustainability has grown too. This raises the question: How do car manufacturing companies integrate sustainable practices and other “green” alternatives to existing processes to satisfy the sustainability requirement?

The aforementioned question is a very broad and wide question, so to split it up into several smaller questions which will be today's topic of discussion: Sustainability in cars & their reusability, sustainability in production & efficiency and sustainable materials & sourcing them

What is sustainability?

Sustainability simply means the maintaining and preservation of the environment, economy and society to prevent destroying them and allow future generations to experience the same things we can now experience and is split into three categories

The planet pillar is sustaining the environment by preserving resources, protecting ecosystems, reducing waste production like carbon footprints and toxic waste and countless other actions that all slowly build up a sustainable environment that will remain stable for years to come. It allows us to have all our needs without compromising the needs of future people.

The profit pillar is maintaining businesses and profit income to allow for financially stable operations that are at little risk of disappearing the long run. It is especially important to car manufacturers to keep development of new technologies active and opens up many more paths to more sustainable processes

The people pillar is the protection of individual and communities' human rights and ensuring fairness and basic human health and need support, it builds a strong society

This is the explanation of the general term sustainability, but looking at it from a global perspective changes everything as every country has their own set of regulations and standards which they label as the baseline for sustainability in manufacturing. In response to that car manufacturers must ensure their products pass all requirements for the markets that the product is intended to be sold in.

Sustainability in cars and their reusability

Now that the main idea of sustainability has been covered, let us look at how the final produced car can be made sustainable. For a car to become “sustainable” there are several key things to consider along with some methods on achieving the desired effect.

The first of which is the cutting down of released emissions to reduce a cars carbon footprint, cars as of today have three main powertrains:

- The BEV (battery electric vehicle) powertrain that relies on batteries to supply energy
- The ICE (internal combustion engine) powertrain that uses fossil fuels like petrol and gasoline
- The HEV & PHEV (hybrid and plug in hybrid electric vehicle) that is a combination of a fuel combustion cell and electrical batteries with the PHEV having larger batteries than the HEV.

The electric vehicles are known as zero emission vehicles because they take energy directly from the batteries and doesn't require a fuel cell so are technically the greenest option if the electricity in the batteries is sourced from sustainable sources like solar or wind farms – but emissions isn't the only part of choosing the most sustainable powertrain. It is equally important to consider the efficiency of it too; however, the efficiency of any powertrain depends on the exact type used, efficient hybrid that relies on lower carbon emission fuels and electricity from clean sources will definitely be superior to the BEV because of another reason: global availability.

When a car is being designed its designers must also think strategically and choose powertrains that use globally available resources. It wouldn't make sense to launch an electric car in a market with little to now available methods of charging the cars batteries – this is the issue present in many countries now.

In addition to the powertrain a car should also have a good exhaust filtering system to further decrease emissions by converting harmful emissions to less damaging ones, a method of doing so is by using a catalytic converter.

With emissions cut down, the next thing to tackle is the reliability and reusability of a car. Thousands of cars if not more reach the end of their life every year, many end up in car dumps to be crushed to metal, which is not very practical, so car manufacturers have recently started using more reusable parts in their vehicles or repurposing old parts to new jobs like taking a cars batteries and using them as energy storage. Even separating parts by material to reuse the raw materials used to originally build the vehicle. This makes a loop of resources that decreases the number of new materials that go into the automotive industry.

Reliability on the other hand, aims to prevent a car from reaching its end of life, by having robust systems and good maintenance to slow down engine wear and car aging as a whole. A reliable car doesn't only depend on production; it is also affected by service and whether its owner cares well for the car and fixes its issues if it gets any and continuously servicing it.

Some ways of producing reliable cars is by having small continuous improvements in a cars' production line and always using tested components that are proven to work perfectly instead of rushing newer components that are not as reliable.

Sustainability in production and efficiency

A lot of a vehicles' emissions are actually released before the vehicle is even fully built, although indirectly, the tools and processes used to build a vehicle release wasteful products which add up across the millions of cars produced yearly, combined these products take a heavy toll on the environment. However, it is not only emissions that affect the sustainability factor, efficiency is also a big part. Having efficient production lines will cut energy and material losses and save time, as well as prioritizing automation which increases consistency and precision across produced units.

While a car is being built, there are several main processes that can be optimized to achieve lower production speed and minimize waste, some of these processes are listed below with their respective enhancements:

- The first step in manufacturing an automobile is to stamp sheets of aluminum into the cars main body panels such as the roof and doors, normally this uses pre-made molds that punch the required shape out, this produces a lot of waste and raises issues once a different panel is needed by the same machine as well as being very inefficient if small tweaks to a panels' design is made. To resolve this computer-controlled lasers are now being used to cut custom shapes from the aluminum sheets in place and reusing all excess metal by melting it back into new sheets to be cut again.

- Engine blocks, transmission blocks can't be manufactured the same way as panels, instead they are drilled out of large blocks of metal that generates a lot of friction and heat which requires massive amounts of cooling-chemicals, so to replace the large amounts of coolant, a technique called micro-lubrication machining is used. Essentially all this does is spray extremely thin layers of biodegradable lubricants onto the tool tip only that vaporizes after use and can be recycled.

- After painting a car, it needs to undergo several layers to cure the paint and spray protective layers to increase longevity and this requires high temperatures that use up a lot of energy, it is like running a large oven at max temperature. To reduce the temperatures needed, special chemicals are added to make the sprayed layers bond and cure much faster at lower temperatures – much more efficient painting.

Although not a direct part of the production line, the logistic methods used to transport raw materials and components throughout a factory also contribute to the overall sustainability of a workplace. Traditionally, fixed conveyor belts and human-operated forklifts are used that introduces human error and slows transport. To reduce error automated guided vehicles (AGVs) that rely on lined out paths or rails to transport items to most places they need to go to.

A few of the earlier stated optimizations create a closed loop of materials, like no excess aluminum or loss of biodegradable oil in engine block milling, this eliminates waste from these sources. By looking back, we can now see that efficiency is linked to sustainability, by increasing efficiency less waste is removed and energy costs are less, an eco-friendlier system that is environmentally sustainable.

Sustainable materials and sourcing them

For the last point of the article, manufacturers seeking more sustainable materials aim for materials that are produced with little to no carbon emissions and part of these materials is the energy used in powering a factory – one of the main resources needed to operate a factory. Some methods have been developed recently to get more “green” materials and more companies are starting to use these methods as regulations get stricter in many countries.

Two of these materials is listed below with their more economical way of acquiring them:

- The metals, steel and aluminum are a key resource in almost all automobiles, making steel and aluminum in traditional ways produces a large number of carbon-containing compounds and recently methods have been found like Hydrogen Direct Reduced Iron for steel which only produce oxygen as a product.

- Plastics, plastics are widely used in interiors and other parts of a vehicle because they are lightweight, relatively cheap and easy to produce in many shapes, but plastics have one flaw, they don't naturally decay, they are non-biodegradable. Bioplastics, as the name suggests, are a type of plastics that can degrade naturally without any external chemical interference but are pricier than regular plastics, nonetheless more manufacturers are moving to bioplastics to further their cars sustainability and ensuring easier recyclability

- Batteries, especially in EV cars nowadays, new rules being passed in some countries are now making batteries fully recyclable by reusing the elements used in batteries like lithium and nickel.

Energy sourcing on the other hand focuses on decarbonizing the energy grid system that provides the energy used in factories, moving from fossil-fuel burning systems to self-sustaining and renewable options such as wind and solar farms. Some manufacturers could even install renewable energy farms directly at or nearby their factories to reduce energy loss by decreasing the distance energy travels.

Yet there is one source of energy loss in a factory that can be taken advantage of, the thermal energy lost by the operating machines in a factory, the painting process discussed earlier may use less energy, but energy loss is still there. A system like an HVAC climate control system could be modified to use the thermal energy emitted to heat other places and maintain constant temperatures across the factory, repurposing what was previously wasted energy.

Sustainability is no longer an optional choice, it is slowly becoming a necessity all over the world, and it does not only affect a particular category, car manufacturers are now developing sustainable strategies constantly and upgrading previous strategies to meet more and more strict requirements. Slowly building the path to a more sustainable future step-by-step – not just for automobiles.

-Abdullah Haider